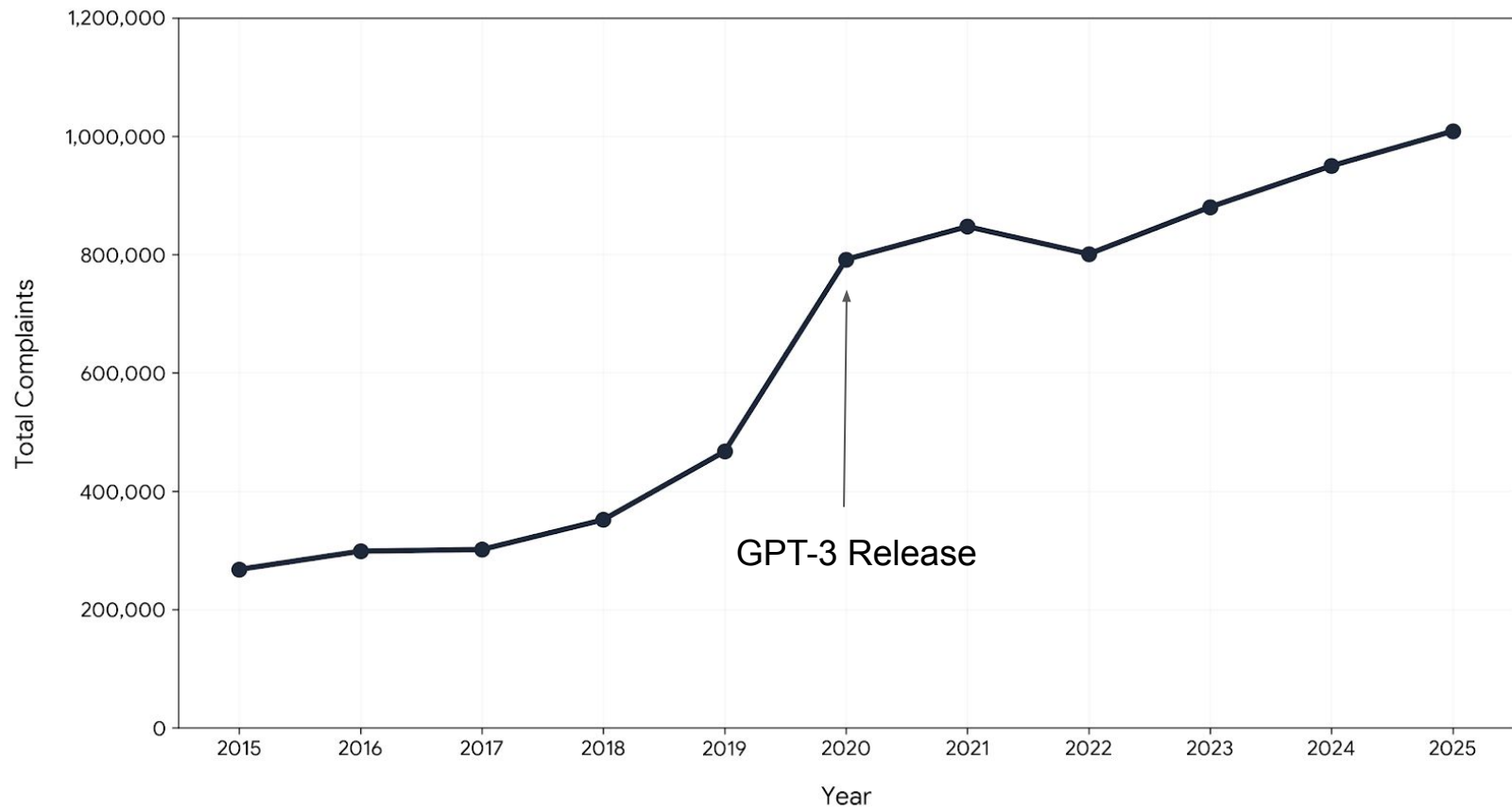


Empathy, Not Telepathy

How Embedded Engineering Context Helps Scale Cyber
Response

Why would I even want
engineering?

The Noise Floor: FBI IC3 Annual Complaint Volume (2015-2025)



Reported Cyber Loss
Has Grown By Over
250%
Since 2020

5x The Rate Of
Cyber Hiring



When everything is a needle, find the sharpest one

Signal Through The Noise

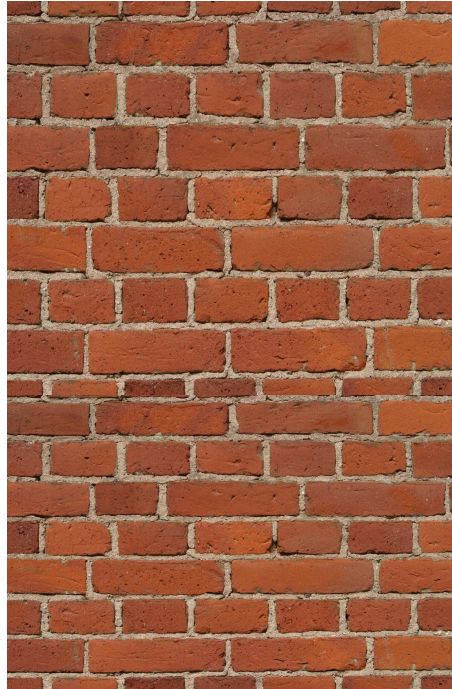
The Context Gap

Ability Context

**Engineering +
Product Context**



**Cyber Operations
Context**



Problem Context

TICKET-RVASEC15: Making a Peanut Butter and Jelly Sandwich

1. Take two slices of bread
2. Put peanut butter on one
3. Put jelly on other one
4. Put them together

TICKET-RVASEC15: Making a Peanut Butter and Jelly Sandwich



Domains Versus Hashes

Should they
have the same
TTL?

Domains Versus Hashes

Should they
have the same
TTL?

‘Verify’

Is this word
ours?

**Domains
Versus
Hashes**

Should they
have the same
TTL?

‘Verify’

Is this word
ours?

0 Hits

Should this be
a relief?

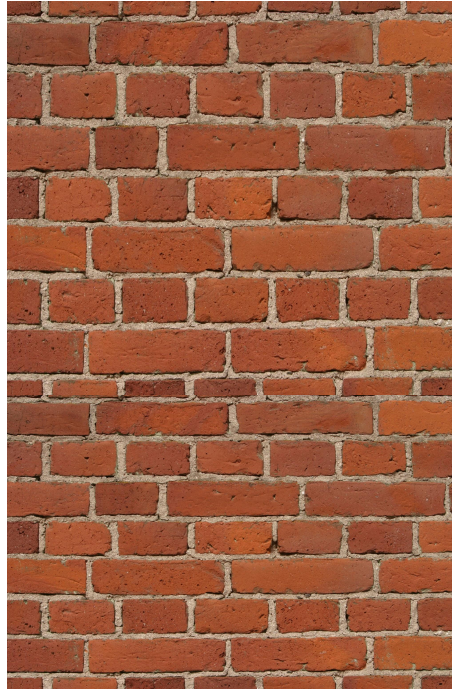
Context

Ability Context

**Engineering +
Product Context**



**Cyber Operations
Context**



Problem Context

Empathy, Not Telepathy

Ability Context

Embedded Context



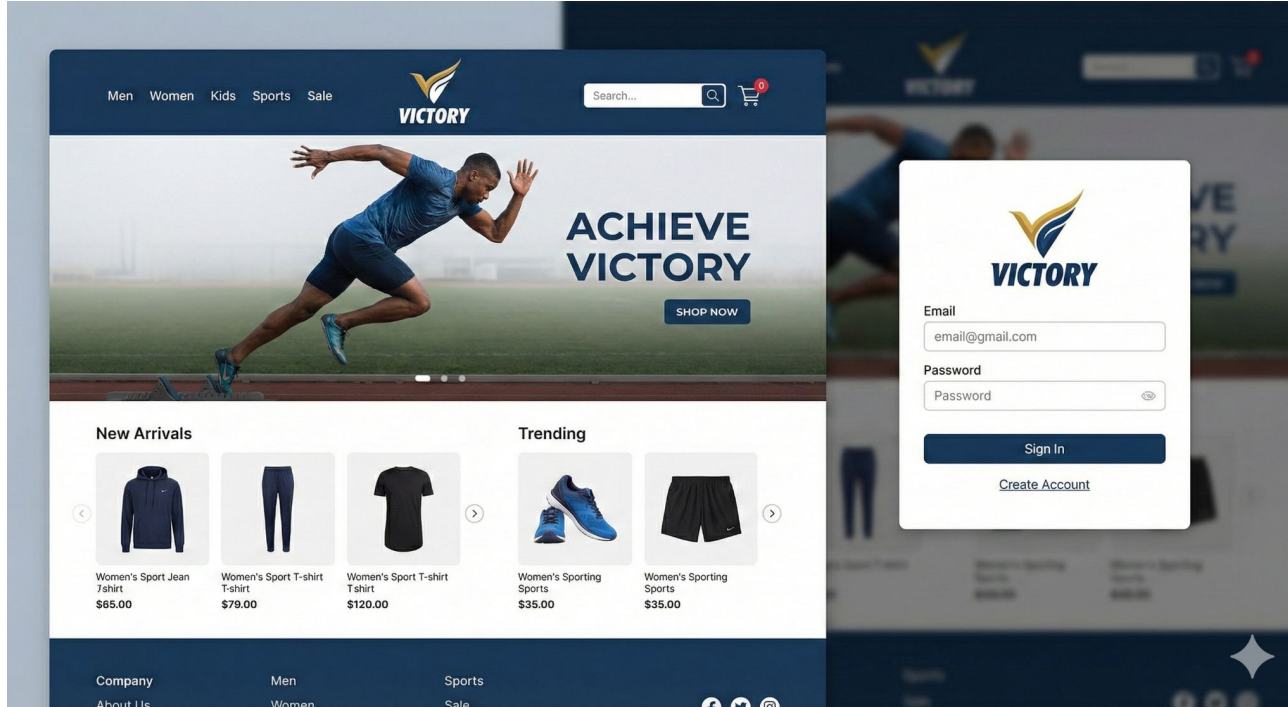
Problem Context

TICKET-RVASEC15: Making a Peanut Butter and Jelly Sandwich



Real World Case Study: Phishing Domains

Fictional Brand Example



AI Will Increase the Quantity — and Quality — of Phishing Scams

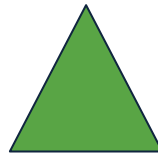
**Consumers lost \$12.5 billion to
fraud last year, and AI-powered
scams are set to explode in 2026**

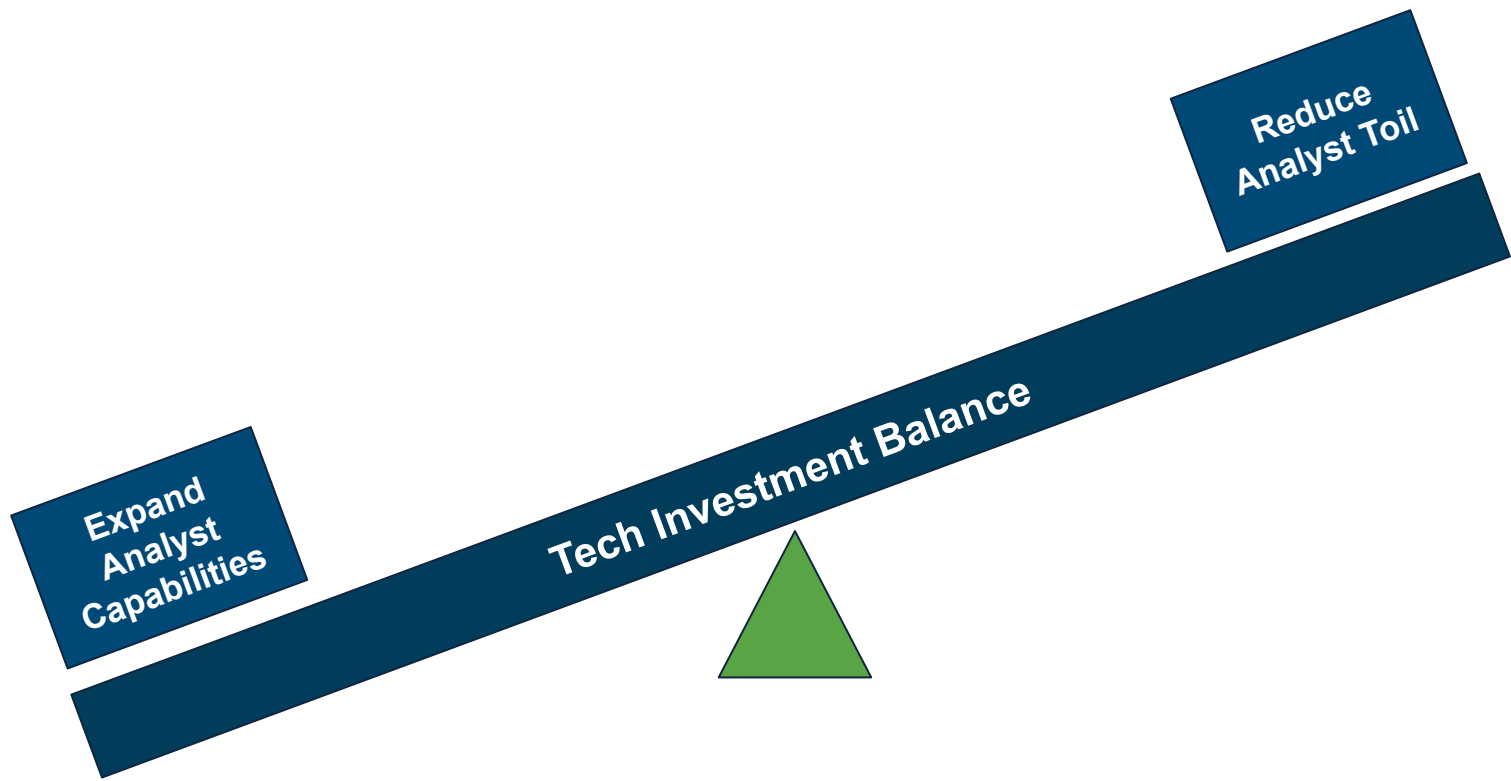
The Balancing Act: Capability vs. Toil

**Expand
Analyst
Capabilities**

**Reduce
Analyst Toil**

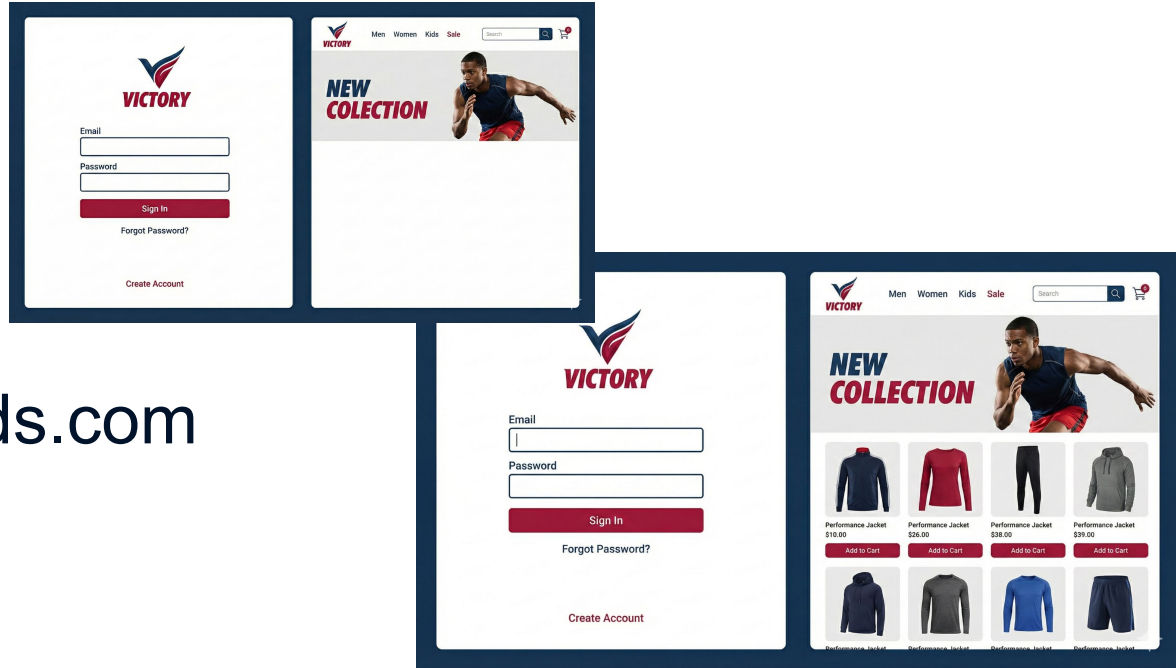
Tech Investment Balance





Typosquatting

- v1ctory.com
- vctory.com
- victorysportinggoods.com

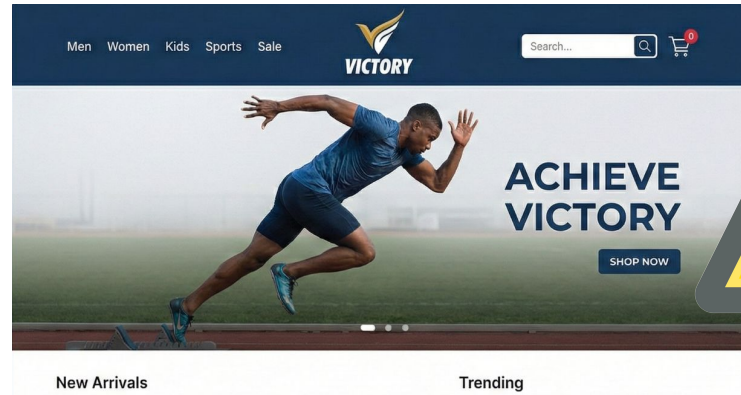
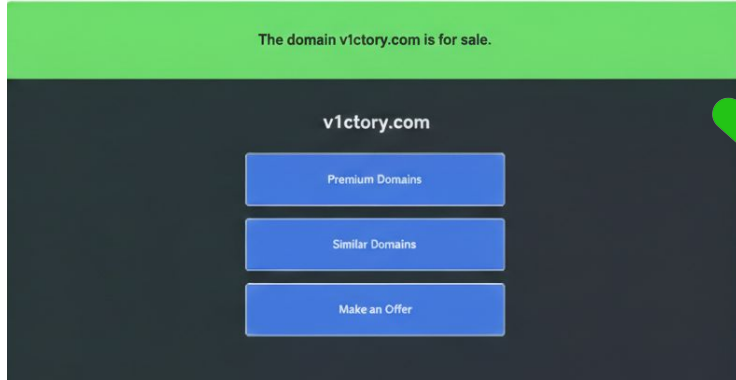


Traditional Response

**Manual
Screenshot
Monitoring**

**Submit for
Takedown**

Manual Monitoring

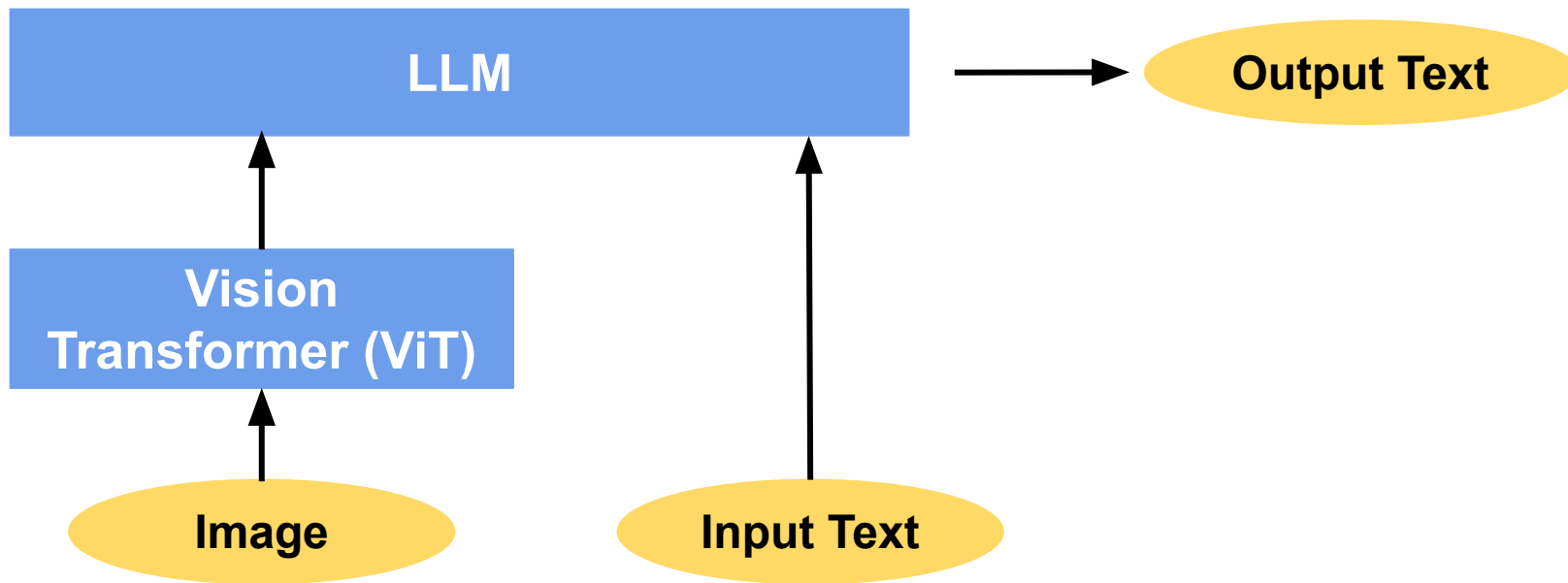


Vision Language Models

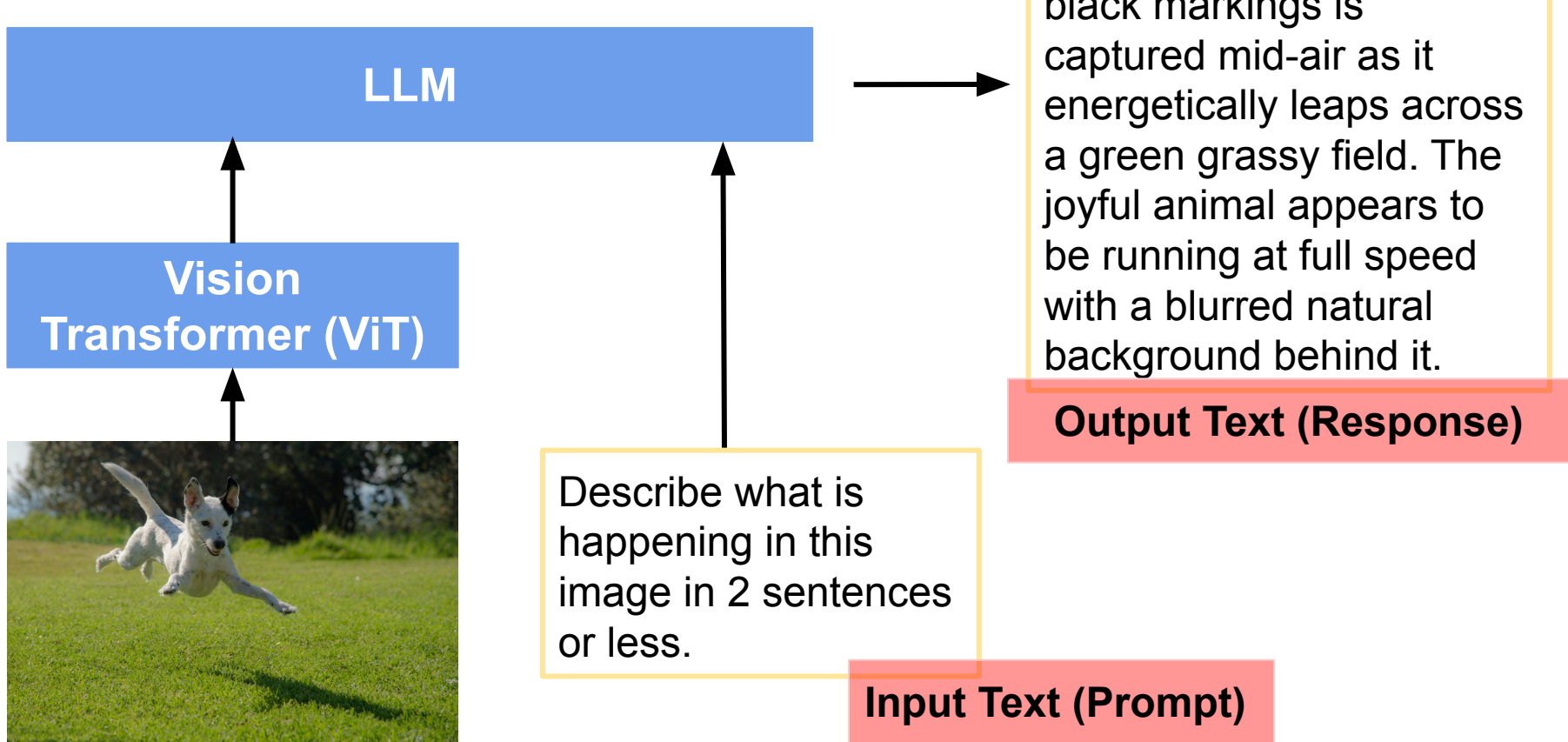
Large Language Model (LLM)



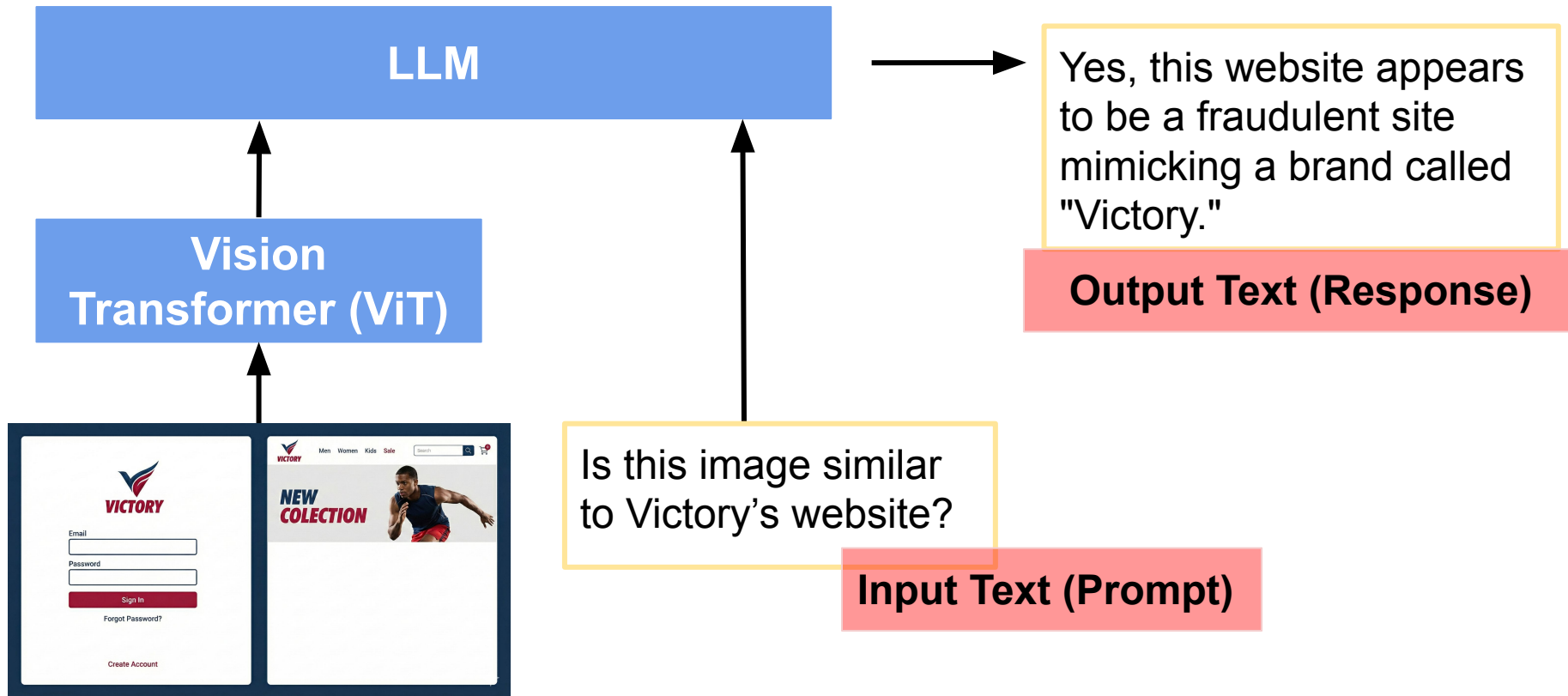
Vision Language Model (VLM)



VLM Demo



Sample Brand Demo



Use VLMs to Detect Phishing

Analyze Screenshot

Screenshot



Live screenshot



Full Image

The domain v1ctory.com is for sale.

v1ctory.com

Premium Domains

Similar Domains

Make an Offer



Analyze Screenshot

Screenshot

Live screenshot

Full Image

About Show

Owner Images Sign Up



Your Dog's Happiest Hours, Guaranteed.

[Our Services](#)

[Meet Our Walkers](#)

[Booking](#)

[Testimonials](#)

[Contact](#)

Ready for a
Happier Dog?
Let's Walk!

[Book a Walk Now](#)

Why Choose Us?



Certified
Walkers



Real-Time
Updates



Happy
Pets

Our Services

Standard Walks

Continue with a
standard walk.
\$3.00

Pack Hikes
(Descriptions
and prices for
your time.
\$12.00

**Specialized
Training**
Descriptions
trainings, &
specialized treat.
\$18.00

What Our Clients Say

"With a whiffy
quotes an a
great happy"



Ronone

"With short
quotes from happy
dog owners and
photos of their
pets."



Martin a

"Se is our
happiest end we
went moms."



Semara

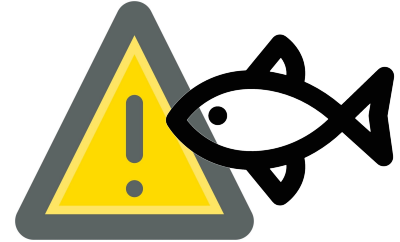
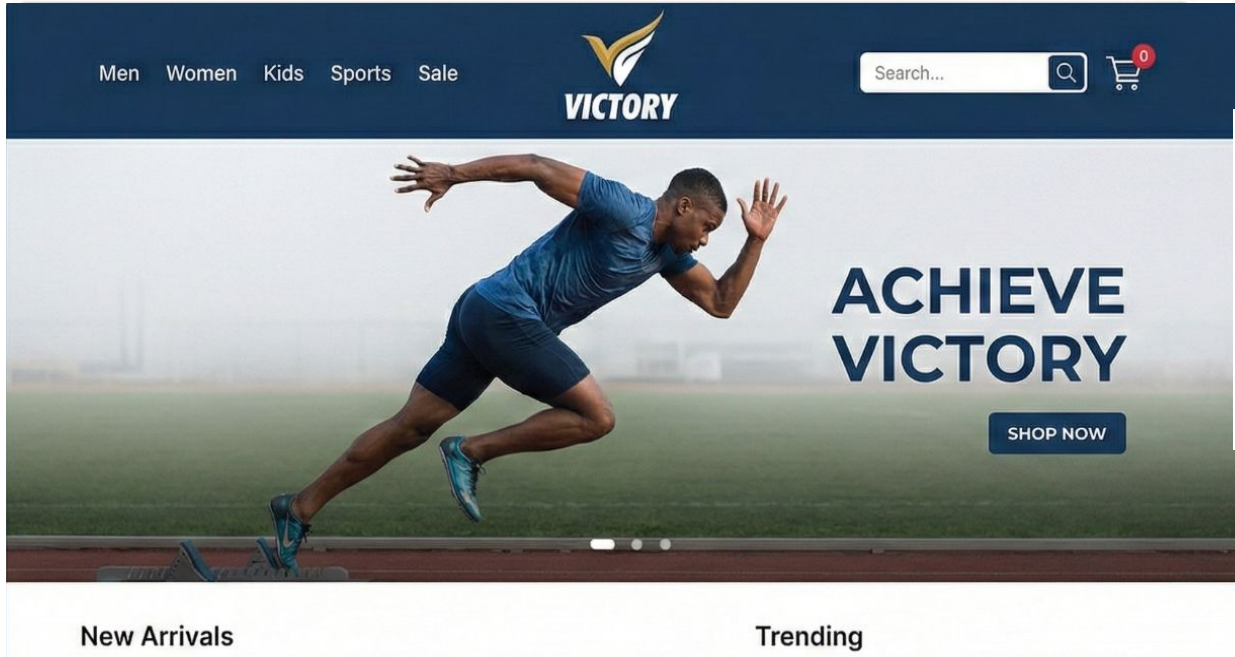


Analyze Screenshot

Screenshot

Live screenshot

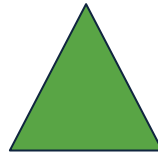
Full Image

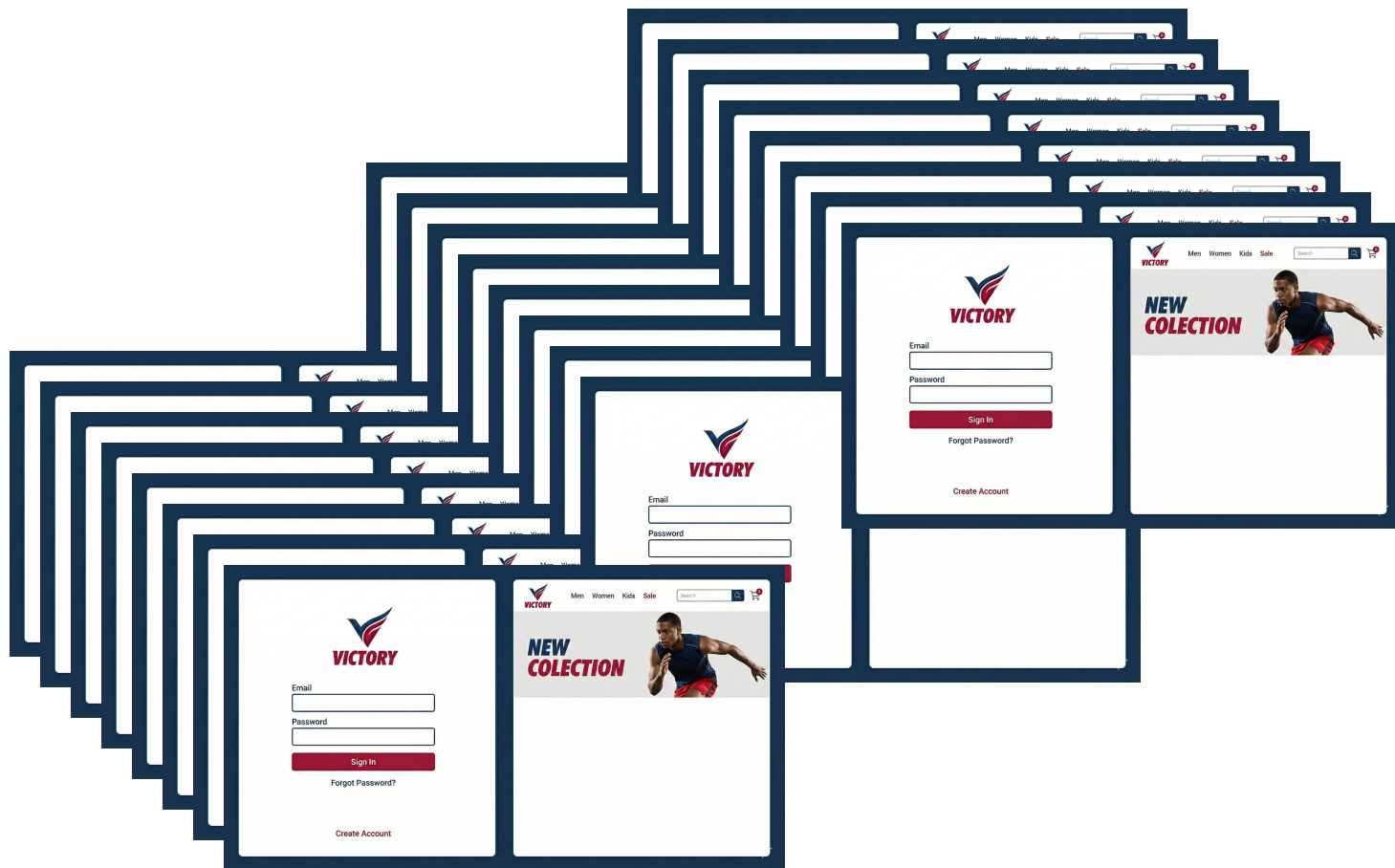


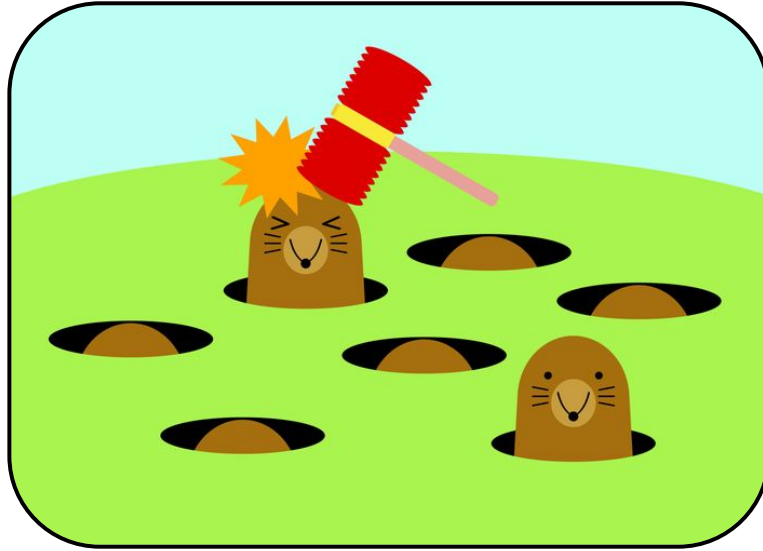
**Expand
Analyst
Capabilities**

**Reduce
Analyst Toil**

Tech Investment Balance







Reactive Response Doesn't Scale

Shift to Large Scale Disruption

Gather

Vendor 1
Data

⋮

Vendor N
Data

Internal
Sources

Data
Pipeline

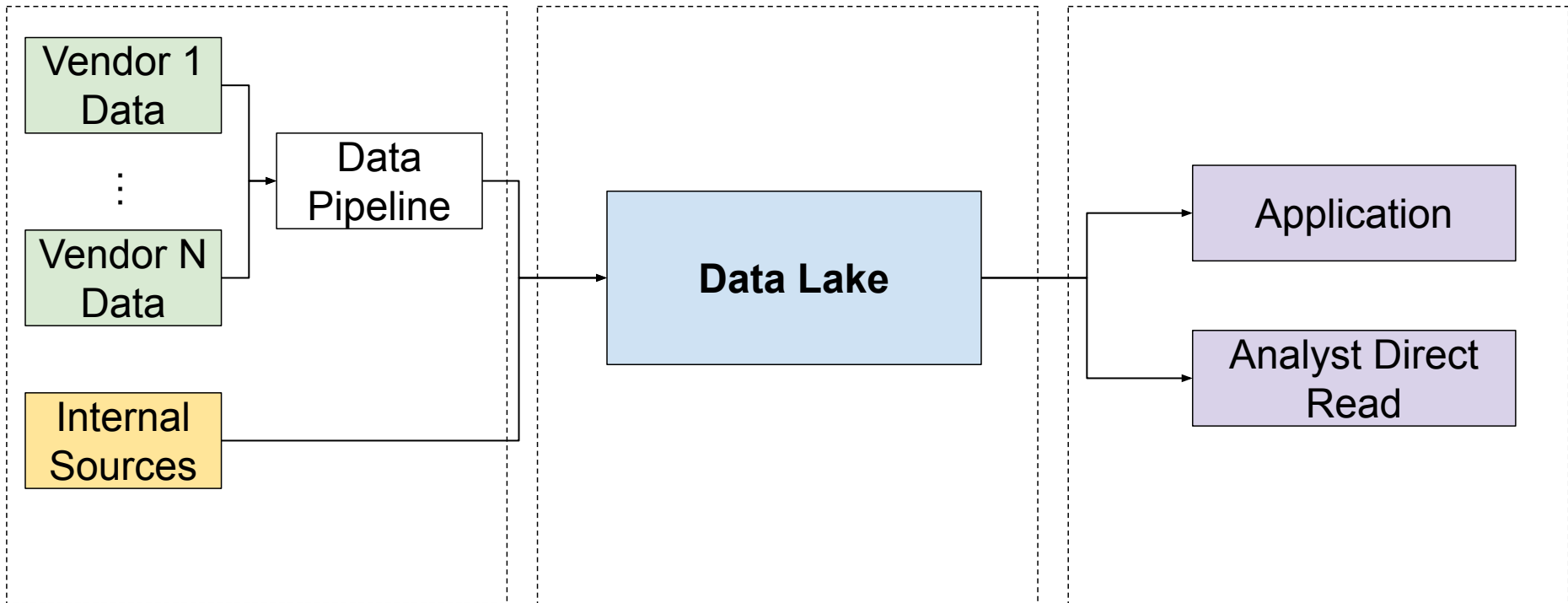
Curate

Data Lake

Disseminate

Application

Analyst Direct
Read

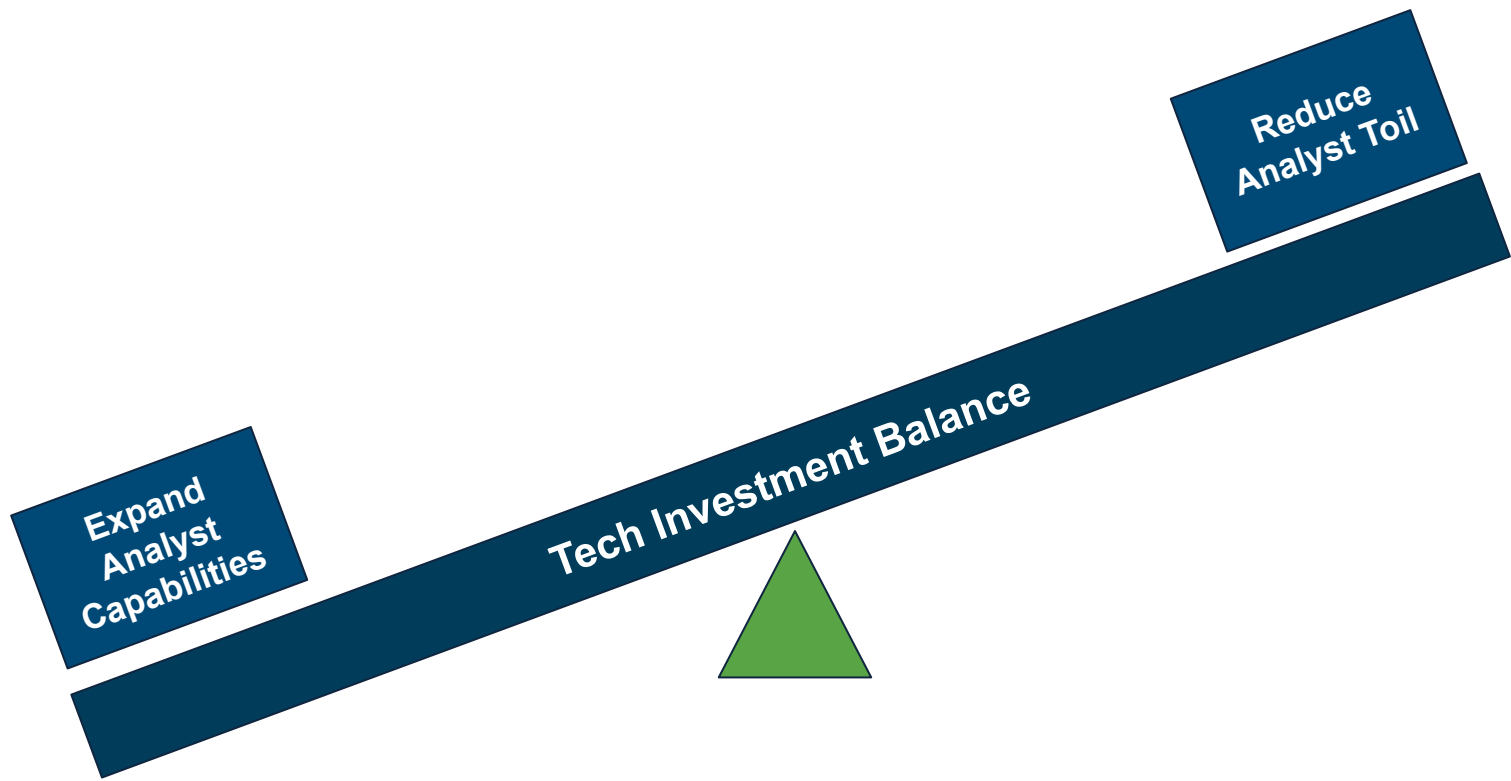




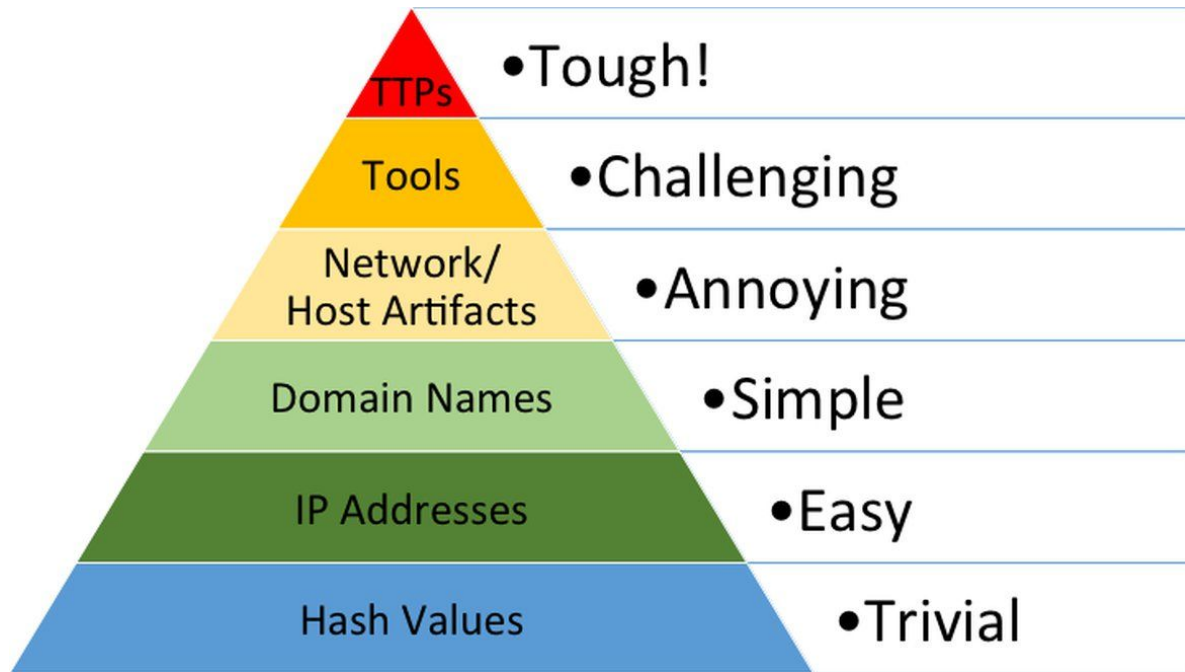
**Data sitting in
vendor tools**

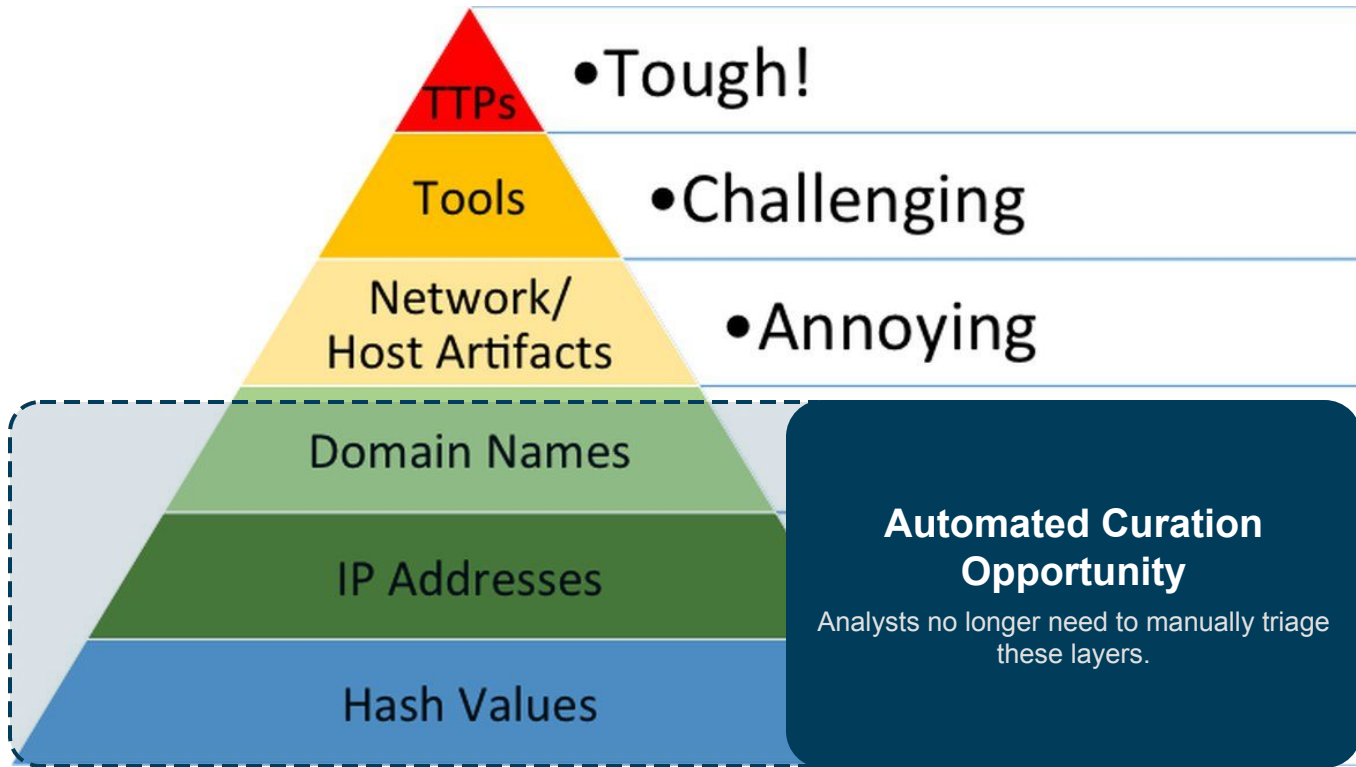


**Data migrated into
the data lake**

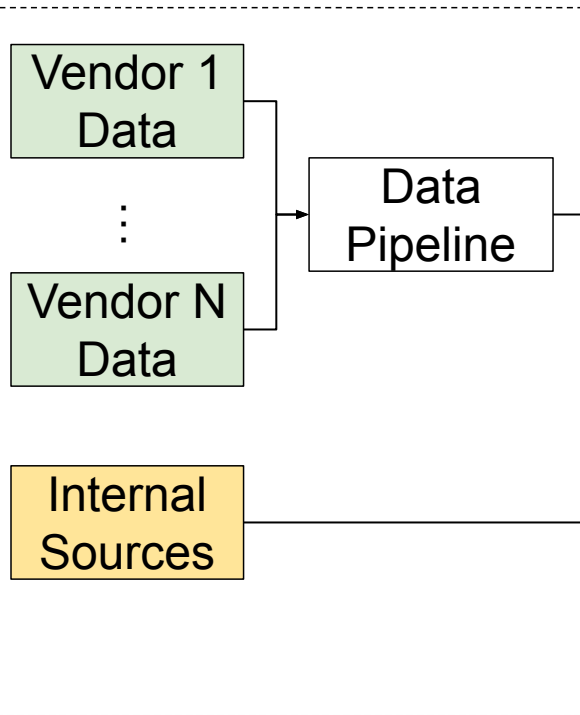


Pyramid of Pain

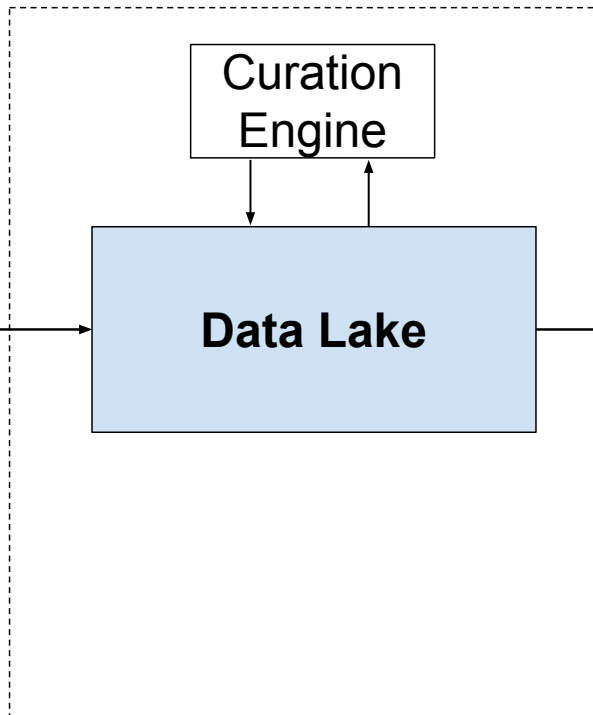




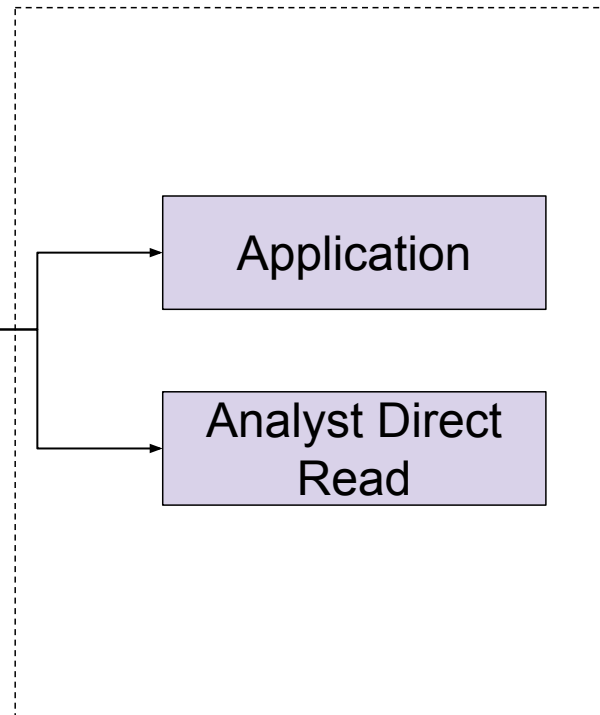
Gather



Curate



Disseminate





**Data sitting in
vendor tools**



**Data migrated into
the data lake**



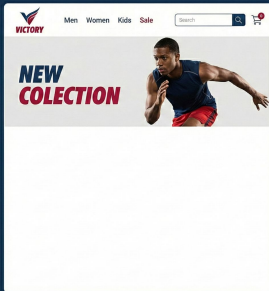
**Curated data provided
to analysts**



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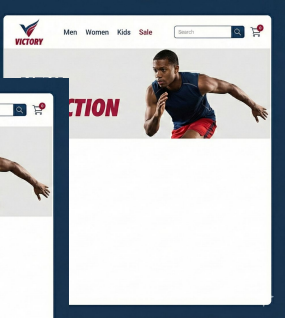
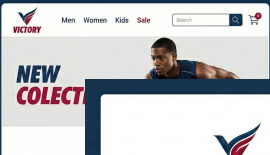




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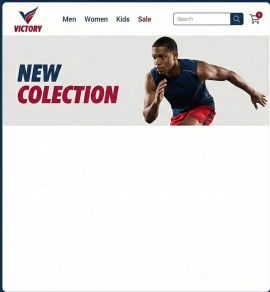




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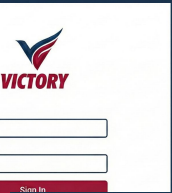
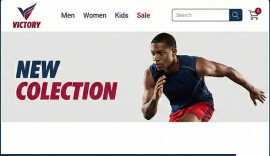




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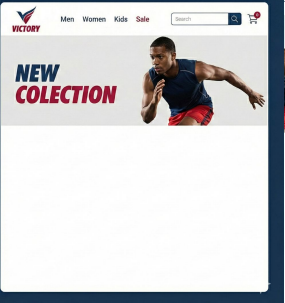
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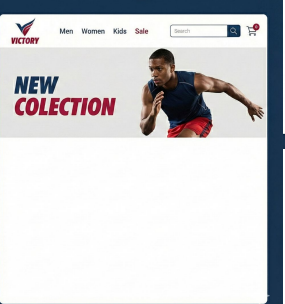




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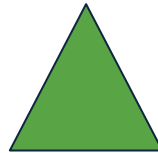


Proactive Response Scales

**Expand Analyst
Capabilities**

**Reduce Analyst
BAU**

Tech Investment Balance



Engineering + Product Context



Empathy

Cyber Operations Context



Takeaways

**Build With,
Not For**

**Build With,
Not For**

**Start Close,
Not Big**

**Build With,
Not For**

**Start Close,
Not Big**

**Momentum
Compounds**

Empathy, Not Telepathy

Empathy, Not Telepathy:

How Embedded Engineering Context
Helps Scale Cyber Response

Questions?

Contacts

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kyle.flaherty@capitalone.com

Connect with me on LinkedIn:
<https://www.linkedin.com/in/kyle-flaherty>